

Behind the [MASK]: Disentangling Representation and Faithfulness in DAPF-Based Dementia Detection

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Abstract

Spoken-language analysis via prompt-based domain-adaptive models is a promising direction for low-resource, non-invasive dementia screening, but such models remain internally opaque. We study the interpretability of the Domain-Adapted models via Prompt-based Fine-tuning (DAPF) framework, which casts dementia detection as diagnosis-related masked-token prediction. We interpret DAPF and strong baselines using a variety of probing and analysis techniques, finding that DAPF achieved the best overall performance (accuracy=0.83 and macro-F₁=0.83) with diagnosis most recoverable from its [MASK] representation. However, this representational advantage did not extend to token-level explanation faithfulness. DAPF attributions primarily reflected language task vocabulary, discourse markers, and transcription artifacts, with perturbation tests showing weak or negative effects. This suggests that its masked-token interface determines diagnosis information without producing faithful token-level explanations.

1 Introduction

Detecting Alzheimer’s disease and related dementias (ADRD) from spoken language offers a low-cost, non-invasive alternative to traditional biomarkers (de la Fuente Garcia et al., 2020; Ding et al., 2024; Lima et al., 2025). Rather than requiring neuroimaging, cerebrospinal fluid collection, plasma biomarkers, or blood-based assays (Khan and Alkon, 2015; Leuzy et al., 2025; Mielke et al., 2024), spoken dialogue can be elicited through picture description, story recall, or conversational interviews (Luz et al., 2020; Lanzi et al., 2023; Clarke et al., 2021; Pope and Davis, 2011). This makes it a useful medium for studying differences in lexical choice, fluency, discourse organization, or other ADRD-associated language changes (Ahmed et al., 2013; Slegers et al., 2018; Gao and He, 2024).

However, cross-corpus generalization is difficult for spoken language ADRD detection. Datasets differ in size, participant demographics, and elicitation tasks, and high accuracy on a specific corpus is not conclusive evidence that a model has learned linguistic or task-specific patterns relevant to ADRD (Farzana and Parde, 2023; Wang et al., 2022). Domain-Adapted models via Prompt-based Fine-tuning (DAPF) frames cross-domain detection as a cloze-style masked-token task by embedding transcripts in prompts containing domain information and diagnosis-related label words (Farzana and Parde, 2024), predicting label words at a [MASK] position and mapping them to actual labels using a verbalizer. From an interpretability perspective, this poses an interesting case. Manual prompts are readable, but not necessarily faithful explanations of a model’s decision (Jacovi and Goldberg, 2020; Lyu et al., 2024). Prompt-based classifiers may also rely on the prompt structure and other dataset-specific cues (Jiang et al., 2020; Hu et al., 2022; Farzana and Parde, 2024).

In this paper, we perform systematic interpretability experiments on DAPF in a controlled cross-domain setting. We use representation probing (Belinkov, 2022) to test whether diagnosis information is recoverable from hidden states at specific positions, and Integrated Gradients (Sundararajan et al., 2017) to assign token-level contributions to the predicted DEMENTIA probability. We aggregate attributions across random seeds by normalized token type, then use deletion and insertion tests (Fong and Vedaldi, 2017; Petsiuk et al., 2018) to evaluate attribution faithfulness and paired error analysis to compare model decisions.

Our findings confirm that many high-attribution tokens reflect the elicitation task or transcription conventions rather than clear ADRD-related language behavior. They also reveal a split between representation-level interpretability and token-level faithfulness. DAPF achieves higher accuracy and

macro-F1 than control baselines, and its final-layer [MASK] representation has more information than its own [CLS] representation and the final-layer [CLS] representations of the baselines. However, perturbation tests show that DAPF’s highlighted tokens do not consistently support the model’s DEMENTIA prediction. Thus, while DAPF appears to concentrate diagnosis information at the prompt prediction position, this does not necessarily lead to faithful word-level explanations.

2 Related Work

2.1 Spoken-Language Dementia Detection

Speech-based ADRD detection has been surveyed extensively (Petti et al., 2020; de la Fuente Garcia et al., 2020; Qi et al., 2023; Ding et al., 2024; Slegers et al., 2018). The most popular dataset for the task is DementiaBank, and the Pitt Corpus within it, based on picture descriptions (Lanzi et al., 2023). ADReSS standardized the Pitt corpus, while ADReSSo and ADReSS-M extended it to speech-only detection and multilingual evaluation (Luz et al., 2020, 2021, 2024). Other previously used datasets include CCC, which contains longer health-oriented conversational interviews (Pope and Davis, 2011), and SLaCAD, which links spoken-language data with clinical and biomarker measures for early ADRD detection (Farzana et al., 2024). Their differences in elicitation task, transcript length, participants, and clinical annotation make them useful for studying cross-domain generalization versus spurious fitting to task-specific vocabulary or dataset artifacts.

2.2 Prompt Learning and Domain Adaptation

Prompt learning has been widely used to adapt pretrained language models through manually designed prompts, automatically generated prompts, and learned continuous prompts (Schick and Schütze, 2021; Shin et al., 2020; Gao et al., 2021; Ding et al., 2022; Li and Liang, 2021; Lester et al., 2021; Liu et al., 2022). It has also been applied to domain adaptation by learning domain-specific prompts while keeping most model parameters fixed (Ben-David et al., 2022; Goswami et al., 2023; Hedderich et al., 2021). In spoken-language ADRD detection, DAPF uses prompt-based masked-token prediction to improve cross-domain performance (Farzana and Parde, 2024). While previous work focused on its predictive performance, we study how the model represents

diagnosis-related information and whether its predictions can be interpreted reliably.

2.3 Representation Probing

Representation probing trains auxiliary classifiers to test what information is recoverable from model’s hidden states (Belinkov, 2022). Layer-wise probing has been used to examine where linguistic information emerges in BERT (Tenney et al., 2019). Since probe performance can reflect the probe’s own learning capacity, prior work recommends simple probes and control tasks (Hewitt and Liang, 2019). We therefore use logistic regression and verify the results with a random-label control.

2.4 Post-hoc Attribution Interpretability

Post-hoc interpretability methods are widely used to identify which parts of an input contribute to a prediction (Madsen et al., 2022). In clinical NLP, a model might appear to perform well based on dataset-specific cues, preprocessing artifacts, or clinically irrelevant tokens (Wang et al., 2022; Vásquez-Venegas et al., 2024). Gradient-based token-level attribution methods estimate the contribution of input tokens from the model’s gradients (Simonyan et al., 2013; Ancona et al., 2019; Smilkov et al., 2017; Sundararajan et al., 2017); for instance, Integrated Gradients traces how the model’s prediction changes as the input moves from a baseline version to the actual example (Sundararajan et al., 2017). Perturbation-based methods, such as occlusion, LIME, SHAP, and feature ablation, instead measure how the prediction changes when parts of the input are removed, masked, or replaced (Zeiler and Fergus, 2014; Ribeiro et al., 2016; Lundberg and Lee, 2017; Li et al., 2016; Fong and Vedaldi, 2017).

Attribution scores should not be treated as faithful explanations by default. Highlighted tokens may look plausible while failing to reflect the model’s actual decision process (Jacovi and Goldberg, 2020; Poerner et al., 2018). If the highlighted tokens are truly important for ADRD prediction, removing them should weaken that prediction, while retaining or restoring them should help preserve or recover it (Fong and Vedaldi, 2017; Petsiuk et al., 2018; DeYoung et al., 2020). Transcript-based models may capture tokens associated with impairment, but they may also exploit cues specific to the task or corpus, such as picture-description content, transcription/disfluency tags, interviewer interactions, or other dataset-specific artifacts (Cum-

mings, 2019; Farzana et al., 2022; Heitz et al., 2024; Farzana and Parde, 2023; Liu et al., 2024). We use Integrated Gradients, cross-seed attribution aggregation, and deletion/insertion tests to examine whether DAPF’s predictions are supported by stable and faithful token-level evidence.

2.5 Interpretability in Dementia Detection

Interpretability is important for building trust in spoken-language ADRD detection systems. Earlier work emphasized hand-crafted acoustic and linguistic features, including lexical, syntactic, semantic, discourse, and fluency measures (Fraser et al., 2015; Mueller et al., 2018; Davis and MacLagan, 2009; Rohanian et al., 2021; Petti et al., 2020; Qi et al., 2023). More recent studies have applied post-hoc explanation methods to ADRD models, identifying clinically plausible linguistic cues while also showing that explanations can be influenced by corpus-specific and preprocessing-related artifacts (Shankar et al., 2025; Vimbi et al., 2024; Iqbal et al., 2024; Oiza-Zapata and Gallardo-Antolín, 2025; Wang et al., 2020; Zhu et al., 2022; Gallardo Antolín et al., 2025; Farzana et al., 2022; Heitz et al., 2024; Farzana and Parde, 2023; Liu et al., 2024). Most prior work has focused on the clinical plausibility or usefulness of explanations. In contrast, we study whether a prompt-based ADRD detector encodes diagnosis information at the prompt prediction position and whether its token-level attributions are stable and faithful.

3 Task and Model Background

3.1 Task

We study binary ADRD detection from spoken-language transcripts using one treatment model (DAPF) and two baselines (BERT-CLS and BERT-CLS+Prompt) to distinguish the effects of prompts and special tokens. The models are described in §3.2 and §3.3. Given the limited availability of spoken-language ADRD data, we evaluate all models in a cross-domain setting using the Carolina Conversations Collection (CCC) as the source domain (Pope and Davis, 2011) and ADReSS as the target domain (Luz et al., 2020). Details regarding those datasets were provided in §2.1.

We use the *ADReSS train* split as the target training domain for CCC → ADReSS adaptation and evaluate on the held-out *ADReSS test* split. Class distributions are shown in Table 1. The small size of both datasets makes interpretability anal-

Dataset Split	Total	AD	HC
Source Train (CCC)	105	57	48
Target Train (ADReSS)	108	54	54
Target Test (ADReSS)	48	24	24

Table 1: Dataset splits. *AD*=Alzheimer’s disease or related dementia; *HC*=Healthy control.

ysis challenging, as token-level explanations can be sensitive to individual participants, repeated task vocabulary, and model initialization.

3.2 Prompt-Based Domain Adaptation

Domain-Adaptive Prompt-Based Fine-Tuning (DAPF) recasts ADRD detection as a prompt-based masked language modeling task rather than conventional sequence classification (Farzana and Parde, 2024). It introduces two manually-designed prompt components: domain-specific prompt text (*D*), which captures corpus-specific characteristics (e.g., CCC or ADReSS); and class-specific prompt text (*C*), which captures diagnosis-specific cues. DAPF fine-tunes a pretrained language model to predict masked label words (e.g., *AD* or *control* in place of [MASK]) inserted into these prompts, allowing the model to better align its representations with both the target task and differences between the source and target domain data, improving its cross-corpus generalization abilities. To ensure close replication, we use `template_id=7` proposed by Farzana and Parde (2024):

Participant’s narration on
{domain}. {transcript} Patient
has diagnosis [MASK].

Following the original work, we used BERT (Devlin et al., 2019) as the backbone model and prediction head. The manual verbalizer maps words predicted in the masked position to class labels:

HEALTHY → {healthy}
DEMENTIA → {dementia, alzheimer’s,
disfluent, disordered}

This makes DAPF particularly interesting for interpretability, since predictions are made from a single prompt token rather than an aggregated [CLS] sequence representation. If prompt learning works as intended, diagnosis-relevant information should be encoded at the [MASK] position. However, the same structure introduces interpretability risks, which we investigate in this study.

3.3 Baselines

To distinguish DAPF-specific explanation behavior from broader limitations of token attribution, we compare DAPF with two BERT sequence classifier baselines with the same data settings.

BERT-CLS Baseline: This is a transcript-only baseline that receives the transcript as input and predicts the class through a standard classification head over the [CLS] representation, functioning as a more conventional sequence classifier:

[CLS] {transcript} [SEP] $\rightarrow$ classifier head $\rightarrow$ {HEALTHY, DEMENTIA}

The baseline shares the same *BERT-base-uncased* encoder as DAPF, with other settings kept constant. It neither includes the DAPF prompt template nor a diagnosis [MASK] token, so we probe [CLS] only.

BERT-CLS+Prompt Baseline: To separate the effects of prompt text and masked-token prediction, BERT-CLS+Prompt uses the same prompt as DAPF. However, unlike DAPF, it does not predict label words at the [MASK] position and does not use a verbalizer. Instead, it is trained using a classification head over [CLS]. Therefore, we treat its [MASK] representation as a prompt-position artifact rather than a meaningful diagnosis signal. This baseline isolates the effect of the prompt alone, while retaining [CLS] classification.

4 Interpretability Methodology

We interpret using representation probing (§4.1), token- (§4.2) and word-level (§4.3) attributions, and perturbation-based faithfulness diagnostics (§4.4). We also report paired error analyses based on output predictions (§6.2).

4.1 Representation Probing

We evaluate whether diagnosis information is encoded in hidden states using representation probing. For DAPF, we save hidden states at the [CLS] (sequence representation) and diagnosis [MASK] position. For BERT-CLS, no diagnosis [MASK] token exists, so we probe [CLS] only. Since BERT-CLS+Prompt contains both [CLS] and the auxiliary [MASK], we probe both, although only [CLS] is used for prediction. For each condition, we train a logistic regression model on frozen hidden states using stratified five-fold cross-validation and macro-F1 scoring to measure how well diagnosis can be recovered from a representation. We also

perform layer-wise probing to study where diagnosis information is recoverable across model layers.

4.2 Token-Level Attribution

To estimate each token’s contribution to the predicted probability of AD/HD, $P(\text{DEMENTIA})$, as our primary attribution method we use Integrated Gradients (Sundararajan et al., 2017). For DAPF, $P(\text{DEMENTIA})$ is derived from masked-token verbalizer scores at [MASK]. For BERT-CLS and BERT-CLS+Prompt, $P(\text{DEMENTIA})$ is extracted from the [CLS] classification head. We report attribution aggregates after removing the prompt-scaffold tokens. A positive attribution score means that the token contributes toward the DEMENTIA class, while a negative score means that the token contributes away from it. To examine whether the qualitative attribution patterns depend on the explanation method, we apply Partition SHAP (Lundberg and Lee, 2017) to DAPF over all 48 test transcripts. We explain $P(\text{DEMENTIA})$ using `max_evals=500`. Since this analysis covers one seed and is not followed by the deletion and insertion tests, we treat it as an attribution-method sensitivity check rather than as a second faithfulness evaluation. Attribution values are not probabilities and should not be interpreted as clinical effect sizes. They are computed for individual token occurrences rather than types, making them input-specific; each token occurrence may receive a different score depending on its context, position, surrounding words, and the model’s prediction.

4.3 Aggregated Token-Level Attribution

Since individual token occurrences can be misleading, we use the same token preprocessing for both attribution methods and aggregate attributions by normalized token type. We compute both the mean signed attribution, indicating whether a token tends to support or oppose the DEMENTIA prediction, and the mean absolute attribution, indicating the overall magnitude of its contribution (see Appendix A for the definitions). To avoid over-interpreting rare tokens, we analyze tokens that appear at least 10 times and in an average of at least five transcripts. As a sensitivity check, we also evaluated a lower threshold of at least five occurrences and presence in at least three transcripts, which produced similar qualitative results. We therefore report results using the stricter threshold.

Component	DAPF-BERT	BERT-CLS	BERT-CLS+Prompt
Input format	Prompt-wrapped transcript	Transcript only	Prompt-wrapped transcript
Prediction interface	MLM/verbalizer at [MASK]	Classifier over [CLS]	Classifier over [CLS]
Template	Manual template ID 7	-	Manual template ID 7
Verbalizer	Manual	-	-

Table 2: Model-specific differences between DAPF-BERT, BERT-CLS, and BERT-CLS+Prompt. All models share the same backbone, data split, training configuration, class weighting, and random seeds (Appendix Table 15).

4.4 Faithfulness Tests

Faithfulness measures whether an explanation reflects the model’s actual decision-making process rather than simply providing a plausible rationale (Jacovi and Goldberg, 2020). We evaluate faithfulness using deletion and insertion tests. In deletion, we mask the highest-ranked tokens and measure the change in the predicted probability of DEMENTIA. In insertion, we begin with those tokens masked, restore them, and measure whether the prediction recovers (we provide formal definitions of these tests in Appendix B).

We use two token rankings. The first ranks tokens positively attributed toward DEMENTIA. The second ranks tokens by absolute attribution, regardless of whether they push the prediction toward or away from DEMENTIA. Since Integrated Gradients produces explanations for each transcript, we evaluate faithfulness at the token-occurrence level.

5 Experimental Setup

5.1 Data, Models, and Training

Table 2 summarizes the controlled comparison. All models use the same backbone, data split, training configuration, class weighting, and random seeds. BERT-CLS compares DAPF with a standard [CLS]-based prediction mechanism. The data and backbone were fixed while the [MASK]-based objective was replaced with a standard classification head. Models are evaluated using balanced accuracy, macro F1, AUROC, and Expected Calibration Error (ECE) (Guo et al., 2017); lower ECE indicates better calibration. Full training and interpretability configurations are reported in Appendix Tables 10 and 15.

6 Results

6.1 Predictive Performance and Calibration

Table 3 reports predictive performance (Appendix Table 11). DAPF achieves the highest accuracy and macro-F1 among the baselines, while BERT-CLS achieves the highest AUROC. ECE is simi-

Model	Acc.	Macro-F1	AUROC	ECE
DAPF	0.8292	0.8287	0.8757	0.1683
BERT-CLS	0.8083	0.8063	0.8906	0.1695
BERT-CLS+Prompt	0.8083	0.8065	0.8819	0.1680

Table 3: Mean test performance over five random seeds.

Correct Model Combination	Count
All three correct	170
All three incorrect	19
DAPF only	20
BERT-CLS only	3
BERT-CLS+Prompt only	2
BERT-CLS + BERT-CLS+Prompt	17
DAPF + BERT-CLS+Prompt	5
DAPF + BERT-CLS	4

Table 4: Paired error analysis over 48 ADReSS test samples $\times$ five random seeds ($n=240$ decisions). Counts show which model combinations were correct.

lar across all models. Since BERT-CLS+Prompt closely matches BERT-CLS in accuracy and macro-F1, adding the DAPF-style prompt does not reproduce DAPF’s behavior. The remaining analyses examine whether DAPF’s masked-token prediction objective changes its diagnosis encoding location and its token attribution faithfulness.

6.2 Paired Error Analysis

We compare predictions at the seed-sample level, using the 48 test samples across five random seeds ($n=240$ decisions). Table 4 summarizes paired-error patterns. All models agree on 170 correct and 19 incorrect cases. DAPF is the only correct model in 20 cases, compared with three for BERT-CLS and two for BERT-CLS+Prompt. However, the classifier baselines are both correct in 17 cases where DAPF is wrong. This suggests a modest but distinct decision-making advantage for DAPF.

Model	Representation	Probe Macro-F1
DAPF	[CLS]	0.7550
DAPF	[MASK]	0.7989
BERT-CLS	[CLS]	0.7704
BERT-CLS+Prompt	[CLS]	0.7683
BERT-CLS+Prompt	auxiliary [MASK]	0.7662

Table 5: Final-layer representation probing. BERT-CLS has no diagnosis [MASK]. For BERT-CLS+Prompt, [MASK] is auxiliary and not used for prediction.

Representation	Diagnosis Probe	Random Probe
DAPF [CLS]	0.7550 $\pm$ 0.0113	0.474 $\pm$ 0.007
DAPF [MASK]	0.7989 $\pm$ 0.0189	0.470 $\pm$ 0.004

Table 6: Random-label control, with macro-F1 averaged across five random seeds. Random probes use the same DAPF representations and probing procedure but randomly permute diagnosis labels.

6.3 Representation Probing

Table 5 reports final-layer probing (Appendix Table 12). DAPF’s [MASK] representation is the most diagnostic final-layer representation, outperforming DAPF [CLS] across seeds, with a mean paired macro-F1 improvement of 0.044 (bootstrap 95% CI: [0.034, 0.056]). A one-sided exact Wilcoxon signed-rank test confirmed that the supervised [MASK] representation contains more diagnosis information than [CLS] ($p=0.031$). BERT-CLS [CLS] is slightly stronger than DAPF [CLS], but weaker than DAPF [MASK]. BERT-CLS+Prompt’s final-layer [CLS] probe score is close to BERT-CLS, while its auxiliary [MASK] representation also contains diagnosis information, despite not being used for prediction. This suggests that the prompt [MASK] token accumulates diagnosis-relevant contextual information, but its auxiliary [MASK] representation is less informative than DAPF’s supervised diagnosis [MASK], indicating that DAPF’s masked-token objective further specializes the prompt prediction position.

As a sanity check, we repeated the DAPF probing after randomly shuffling diagnosis labels, while keeping the hidden representations and class balance unchanged. Probe macro-F1 performance dropped from 0.755 to 0.474 for [CLS] and from 0.799 to 0.470 for [MASK] (Table 6), indicating that the probes recover diagnosis-related rather than arbitrary structure.

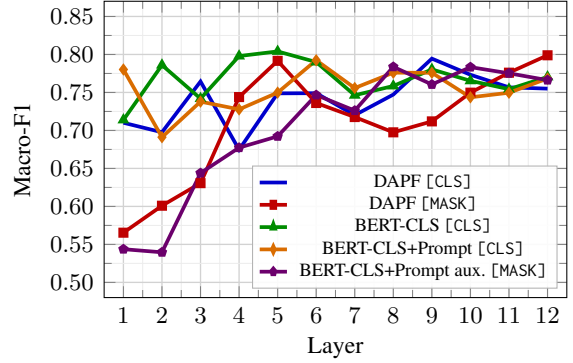


Figure 1: Layer-wise probing across models. Macro-F1 is averaged across five random seeds.

6.4 Layer-wise Representation Probing

Figure 1 shows layer-wise probing results (Appendix Table 13). DAPF’s [MASK] representation becomes informative in the middle layers and peaks in the final layer. BERT-CLS [CLS] peaks around layer 5, suggesting that middle-layer representations contain useful diagnosis information, while DAPF concentrates it at the prompt prediction position.¹ The auxiliary [MASK] in BERT-CLS+Prompt is weak in early layers but becomes more diagnosis-informative after incorporating transcript context in subsequent layers, reaching its strongest values around layers 8–10 rather than the final layer. Thus, it appears that the prompt-as-input setting can produce a diagnostically informative auxiliary prompt position while supervised masked-token prediction further specializes the final [MASK] representation.

6.5 DAPF Attribution Patterns and Method Sensitivity

Across five seeds, 163 normalized token types met the frequency threshold. Signed attributions were centered near zero, mean absolute attribution was modest, and positive and negative tokens were nearly balanced (see Appendix Table 14 for specific values). Table 7 shows the highest positive-attribution tokens. These include Cookie Theft vocabulary (e.g., reach, cookies) together with generic discourse words (e.g., something, well), suggesting that DAPF emphasizes task-specific vocabulary and conversational markers rather than clearly interpretable ADRD-related features.

¹Layer 4-5 results should be read cautiously, since they may encode lexical or task-related differences between groups; probing does not reveal what drives the separation. Future work could investigate random-label controls and probe nuisance variables (e.g., transcript length or number of task-related content words).

Token	Total	T/S	Attr.	SD	Attr.
reach	55	8	0.2540	0.1900	0.3374
out	240	32	0.1832	0.1569	0.2337
something	25	5	0.1827	0.0946	0.2568
anything	40	6	0.1262	0.0808	0.1584
cookies	150	21	0.1130	0.1906	0.2982
quiet	25	5	0.0921	0.1086	0.1211
well	145	21	0.0717	0.0298	0.1149
nice	35	5	0.0657	0.0312	0.0915
no	45	7	0.0630	0.0486	0.0860
by	25	5	0.0585	0.0553	0.0799

Table 7: Top positive DAPF cross-seed mean attribution tokens. Positive values indicate average contribution toward the DEMENTIA class. *Total*=total occurrences; *T/S*=mean transcripts per seed; *Attr.*=mean signed attribution; *SD*=standard deviation across seeds; *|Attr. |*=mean absolute attribution.

Token	Total	T/S	Attr.	SD	Attr.
okay	60	12	-0.5181	0.3719	0.5775
lid	40	7	-0.1401	0.0474	0.1456
guess	45	8	-0.1141	0.0574	0.1407
cabinets	35	5	-0.1017	0.0865	0.1366
curtains	55	11	-0.0977	0.0196	0.1726
laughing	35	7	-0.0813	0.0523	0.1502
washing	55	9	-0.0780	0.0283	0.1133
uh	635	41	-0.0775	0.0240	0.1171
off	145	20	-0.0665	0.0303	0.0997
xx	95	11	-0.0665	0.0547	0.1555

Table 8: Top negative DAPF cross-seed mean attribution tokens. Negative values indicate average contribution away from the DEMENTIA class. *Total*=total occurrences; *T/S*=mean transcripts per seed; *Attr.*=mean signed attribution; *SD*=standard deviation across seeds; *|Attr. |*=mean absolute attribution.

Table 8 shows a similar pattern, including task vocabulary, discourse words, fillers, and transcription artifacts. Although fillers can be associated with cognitive impairment (Farzana et al., 2022), attribution alone does not justify interpreting uh as evidence against ADRD. Overall, the aggregated attributions are more stable across seeds but remain clinically difficult to interpret.

Partition SHAP produced a similar overall pattern. High-attribution items primarily included Cookie Theft vocabulary, such as *fall*, *reach*, *cookie*, and *dish*; discourse forms, such as *well*, *oh*, and *but*; and longer token fragments, such as *ing* and *tin*. Agreement on individual tokens was only partial. Among the 20 tokens reported in the Integrated Gradients top-positive and top-negative tables, 11 (55%) had the same mean direction under both methods, with a descriptive Spearman correlation of $\rho = 0.362$. Overall, the two methods

Model	Test	5%	20%	50%
DAPF	Deletion	0.0080	0.0055	-0.0318
DAPF	Insertion	-0.0187	-0.0352	-0.0916
BERT-CLS	Deletion	0.0259	0.0142	0.0067
BERT-CLS	Insertion	0.0369	0.0813	0.1223
BERT-CLS +Prompt	Deletion	0.0302	0.0426	0.0259
BERT-CLS +Prompt	Insertion	0.0279	0.0787	0.1225

Table 9: Positive-ranking faithfulness comparison using tokens ranked by positive attribution toward DEMENTIA. Positive values indicate behavior consistent with dementia-supporting evidence. Results are shown for 5%, 20%, and 50% perturbation fractions.

highlight similar types of cues, but do not consistently agree on which words are most important.

6.6 Faithfulness Comparison

Table 9 compares perturbation-based faithfulness for tokens ranked by positive attribution toward DEMENTIA (Appendix Table 18). DAPF shows weak deletion effects and consistently negative insertion effects. In contrast, both BERT-CLS and BERT-CLS+Prompt exhibit positive deletion and steadily increasing insertion scores.

Although DAPF has the strongest final-layer diagnosis representation, its positively attributed tokens do not recover the predicted probability of DEMENTIA under insertion. In contrast, BERT-CLS shows more responsive positive attributions under insertion. Likewise, BERT-CLS+Prompt shows positive insertion effects, suggesting that prompt text alone is not responsible for DAPF’s weak insertion faithfulness. The masked-token/verbalizer prediction setup or the attribution behavior when the decision depends on the prompt prediction position are likelier culprits.

We also evaluated faithfulness using tokens ranked by absolute attribution and found that the overall pattern was similar (Appendix I). Overall, DAPF combines a strongly diagnostic supervised [MASK] representation with weak positive-attribution faithfulness. Since BERT-CLS+Prompt does not show the same insertion pattern, the difference is unlikely to result from prompt text alone.

7 Discussion

Representation-level encoding and token-level faithfulness diverge. Classification and probing results show that DAPF learned meaningful

diagnosis-related information, with predictive performance slightly higher than BERT-CLS. The final-layer [MASK] representation was consistently more diagnosis-informative than DAPF’s [CLS] representation, supporting the hypothesis that diagnosis information is concentrated at the supervised prompt prediction position. However, faithfulness tests show that BERT-CLS positive attributions are more responsive under insertion than DAPF’s, uncovering a mismatch between representation-level diagnosis encoding and token-level faithfulness.

Why are DAPF attributions less convincing?

Predicting ADRD through the prompt [MASK] position combines information from the transcript, prompt wording, domain phrase, and verbalizer words, creating challenges in explaining why positive attributions fail under insertion. One possibility is that these interactions are not well localized to individual words. Partition SHAP reproduced the same broad cue categories but agreed only partially with Integrated Gradients on the direction and ranking of individual tokens. This does not by itself establish unfaithfulness, but it suggests that the word-level explanation is sensitive to the attribution method. BERT-CLS has a simpler prediction path, mapping [CLS] representations directly to class logits, which may better align its positive attributions with the insertion test. BERT-CLS+Prompt uses the same prompt-wrapped input as DAPF but predicts through a [CLS] head, resulting in behavior closer to BERT-CLS in aggregate performance and paired-error patterns and indicating that DAPF’s differences are not explained by prompt text alone. Further controls could isolate the roles of the masked-language-model head, the manual verbalizer, and the task/domain prompt.

Prompt-as-input control. BERT-CLS+Prompt behaved more like BERT-CLS than DAPF in predictive performance and attribution faithfulness, while its auxiliary [MASK] representation was less diagnostic than DAPF’s supervised [MASK]. Therefore, prompt wording alone does not explain DAPF’s representation or faithfulness patterns.

Implications for ADRD detection. Future interpretability work should look beyond token attributions to broader transcript patterns, such as informativeness, lexical specificity, and discourse organization. It could also test targeted changes to transcripts based on clinically meaningful features rather than high-attribution tokens. Concept-

based methods could help determine whether these patterns are encoded in the model’s hidden states. More broadly, these findings suggest that interpretability should be considered alongside performance when evaluating low-resource prompt-based models, since strong performance alone does not guarantee faithful explanations.

8 Conclusion

In this paper, we investigated the interpretability of a BERT-based DAPF model for ADRD detection against transcript-only (BERT-CLS) and prompt-as-input (BERT-CLS+Prompt) baselines. While DAPF achieves slightly higher accuracy and unique correct decisions by concentrating diagnosis information at its supervised [MASK] representation, our analyses show that representation-level encoding and token-level faithfulness diverge. Cross-seed Integrated Gradients showed that high-attribution tokens primarily reflect task-specific vocabulary, discourse markers, and transcription artifacts rather than clinically meaningful ADRD markers. Partition SHAP sensitivity check recovered the same broad cue categories, but showed only partial agreement with Integrated Gradients on the direction and ranking of individual tokens. Furthermore, perturbation tests showed that DAPF’s token attributions are less faithful under deletion and insertion than the classifier baselines, indicating that masked-token prompting does not automatically guarantee attribution validity. Ultimately, these findings highlight a mismatch between diagnosis localization and explanation reliability, cautioning that prompt-based model explanations should not be interpreted as faithful clinical rationales.

Limitations

All experiments use a single low-resource CCC → ADReSS transfer setting, so the findings should not be taken as general claims about speech-based ADRD detection. Since masking large portions of a transcript may create out-of-distribution inputs, effects at larger perturbation fractions, particularly 50%, may reflect both token importance and sensitivity to distribution shift (Hase et al., 2021). In the BERT-CLS+Prompt control, we test one DAPF-style prompt; it is possible that different prompt templates, domain descriptions, verbalizers, or prompt-tuning methods may produce different representation and attribution patterns. Finally, representation-level interpretability does not nec-

essarily mean faithful word-level explanations. In DAPF, the [MASK] representation contains strong diagnosis information, but the token attributions do not behave like word-level evidence under perturbation. This limits how strongly the highlighted tokens can be interpreted as explanations for the model’s decisions.

Ethical Considerations

Our experiments use the Carolina Conversations Collection (CCC) and ADReSS datasets, both of which were obtained through their established data access procedures. Access to CCC required institutional ethics review at both our institution and the institution hosting the dataset,² which we obtained. Access to ADReSS required approval through the DementiaBank consortium;³ following that, we additionally obtained an ethics determination at our own institution before using the data. All experiments and reporting of results comply with the datasets’ access agreements and terms of use.

It’s also worth mentioning that these models are intended for research rather than clinical diagnosis. Token attributions that reflect task-specific vocabulary or transcription artifacts may be misleading if interpreted as clinical evidence, while differences across datasets and participant populations may limit generalization.

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²carolinaconversations.musc.edu/ccs/about/

³talkbank.org/dementia/ADReSS-2020/index.html

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A Aggregation of Token-Level Attribution

For each normalized token type w , let $a_i(w)$ denote the attribution assigned to its i -th occurrence. We compute the mean signed attribution

$$\bar{a}(w) = \frac{1}{N_w} \sum_{i=1}^{N_w} a_i(w)$$

and the mean absolute attribution

$$|\bar{a}|(w) = \frac{1}{N_w} \sum_{i=1}^{N_w} |a_i(w)|.$$

The mean signed attribution indicates whether a token tends to support or oppose the DEMENTIA prediction, whereas the mean absolute attribution measures the average magnitude of its contribution regardless of direction.

B Faithfulness Metrics

For deletion, we mask the selected high-attribution tokens and compute

$$\Delta_{\text{del}} = P_{\text{original}} - P_{\text{mask}},$$

where P_{original} is the original predicted probability of DEMENTIA and P_{mask} is the probability after masking the selected tokens. If positively attributed tokens provide faithful evidence for the prediction, Δ_{del} should be positive.

For insertion, we begin from an input in which the selected tokens are masked and compute

$$\Delta_{\text{ins}} = P_{\text{restored}} - P_{\text{masked start}},$$

where $P_{\text{masked start}}$ is the predicted probability for the masked input and P_{restored} is the probability after restoring the selected tokens. Faithful positively attributed tokens should produce a positive Δ_{ins} .

B.1 Interpretability Configuration

Table 10 summarizes the interpretability settings. Representation probing tests whether diagnosis can be recovered from frozen hidden states, while Integrated Gradients assigns token-level contributions to $P(\text{DEMENTIA})$. Since attribution scores are assigned to individual token occurrences, we aggregate normalized token types to identify patterns that are more stable across seeds. Faithfulness is then evaluated with deletion and insertion tests using positive and absolute attribution rankings. We

Analysis	Configuration
Representation probing	Logistic regression probe
Probe evaluation	Stratified 5-fold CV
Probe metric	Macro-F1
Primary attribution method	Integrated Gradients
Attribution sensitivity check	Partition SHAP (DAPF seed 0)
Attribution target	$P(\text{DEMENTIA})$
Attribution unit	Token occurrence
Aggregation unit	Normalized token type
Main aggregation threshold	Count ≥ 10 , samples ≥ 5
Faithfulness rankings	Positive, absolute
Perturbation fractions	5%, 10%, 20%, 30%, 50%

Table 10: Interpretability configuration.

Model	Acc.	Macro-F1	AUROC	ECE
DAPF	0.8292 ± 0.0174	0.8287 ± 0.0173	0.8757 ± 0.0183	0.1683 ± 0.0284
BERT-CLS	0.8083 ± 0.0243	0.8063 ± 0.0247	0.8906 ± 0.0102	0.1695 ± 0.0349
BERT-CLS+Prompt	0.8083 ± 0.0083	0.8065 ± 0.0082	0.8819 ± 0.0156	0.1680 ± 0.0476

Table 11: Mean test performance over five random seeds.

evaluate multiple perturbation fractions to compare local effects with behavior under stronger perturbation; the 50% condition is included as a stress test rather than a precise estimate of token importance.

C Mean test performance over five random seeds with Standard Deviations

Table 11 extends Table 3 by reporting the mean performance together with standard deviations across the five random seeds.

D Final-layer Representation Probing Results With Standard Deviation

Table 12 extends Table 5 by reporting standard deviations across the five random seeds.

E Full Layer-Wise Representation Probing Results

In Table 13 we report full layer-wise diagnosis probing results across DAPF, transcript-only BERT-CLS, and prompt-as-input BERT-CLS+Prompt. Values are macro-F1 scores averaged across five random seeds. For DAPF, [MASK] is the supervised masked-token diagnosis prediction position. For BERT-CLS+Prompt, the auxiliary [MASK] token is present in the input but is not the supervised pre-

Model	Representation	Probe Macro-F1
DAPF	[CLS]	0.7550 ± 0.0113
DAPF	[MASK]	0.7989 ± 0.0189
BERT-CLS	[CLS]	0.7704 ± 0.0260
BERT-CLS+Prompt	[CLS]	0.7683 ± 0.0152
BERT-CLS+Prompt	auxiliary [MASK]	0.7662 ± 0.0374

Table 12: Final-layer representation probing. BERT-CLS has no diagnosis [MASK]. For BERT-CLS+Prompt, [MASK] is auxiliary and not used for prediction.

diction site; prediction is made through a standard [CLS] classification head.

F Summary statistics for aggregated DAPF token attributions

Table 14 shows statistics for aggregated DAPF token attributions.

G Experimental Configuration

Table 15 summarizes the experimental settings shared across all three models: DAPF-BERT, BERT-CLS, and BERT-CLS+Prompt. The only differences between models are the input format, prompt components, and prediction interface, which are described in Table 2 of the main text.

G.1 Computational Resources

All experiments were run locally on a MacBook Pro with an Apple M3 Pro chip, a 12-core CPU, an 18-core integrated GPU, and 36 GB of unified memory. PyTorch operations used the Metal Performance Shaders (MPS) backend where GPU acceleration was supported. All three models use BERT-base-uncased, which has approximately 110 million parameters. The complete experimental pipeline, including training across five random seeds, representation probing, Integrated Gradients, Partition SHAP, and deletion and insertion tests, took approximately 2.5 hours on this device.

H Partition SHAP Sensitivity Analysis

As a sensitivity analysis, we applied Partition SHAP to the DAPF seed whose predictive performance was closest to the five-seed mean. Following the same evaluation setting used throughout the paper, explanations were generated for the 48

ADReSS test transcripts with respect to the predicted probability of the Dementia class.

Partition SHAP was implemented using SHAP’s PartitionExplainer with a maximum evaluation budget of 500 model evaluations per transcript. Unlike the Integrated Gradients analysis in the main paper, this experiment was performed on a single model seed and was intended only to examine whether the qualitative attribution patterns were consistent across attribution methods.

Although the two attribution methods emphasized similar categories of cues, agreement on individual token importance was only moderate. This suggests that the broader qualitative interpretation is relatively stable, whereas precise word-level importance depends on the attribution method. Details can be seen in Table 16.

To examine the cues emphasized by Partition SHAP, we first applied the same token-reporting and normalization procedure used for Integrated Gradients. Partition SHAP itself was computed on the original, unfiltered prompt-wrapped inputs; filtering was applied only after attribution computation and before token-type aggregation. We excluded prompt-scaffold and special tokens, punctuation-only and control symbols, the artifact *ex*, WordPiece continuations containing one or two characters, and isolated single-letter alphabetic fragments other than *i* and *a*. The remaining tokens were stripped of surrounding whitespace, lowercased, and had WordPiece continuation markers removed, while longer subword fragments were retained.

We then aggregated the resulting attribution scores across the 48 ADReSS test transcripts. For each normalized token type, we calculated its total frequency, the number of transcripts in which it appeared, its mean signed SHAP value, and its mean absolute SHAP value. To reduce the influence of rare tokens, Table 17 includes only tokens that occurred at least 10 times and in at least five transcripts. Positive values indicate that a token contributed, on average, toward the DEMENTIA output, whereas negative values indicate a contribution away from it.

The positive and negative rankings contain a mixture of picture-description vocabulary, discourse forms, and tokenization fragments. Words such as *dry*, *fall*, *reach*, *cookie*, *drying*, and *standing* are directly related to the Cookie Theft task. Discourse forms such as *oh*, *well*, *yeah*, and *guess* also receive relatively large attributions, while longer

Layer	DAPF [CLS]	DAPF [MASK]	BERT-CLS [CLS]	BERT-CLS+Prompt [CLS]	BERT-CLS+Prompt aux. [MASK]
1	0.7101 $\pm$ 0.0102	0.5654 $\pm$ 0.0107	0.7138 $\pm$ 0.0148	0.7802 $\pm$ 0.0283	0.5437 $\pm$ 0.0093
2	0.6976 $\pm$ 0.0216	0.6009 $\pm$ 0.0183	0.7860 $\pm$ 0.0130	0.6911 $\pm$ 0.0091	0.5396 $\pm$ 0.0143
3	0.7644 $\pm$ 0.0116	0.6306 $\pm$ 0.0266	0.7419 $\pm$ 0.0142	0.7379 $\pm$ 0.0405	0.6439 $\pm$ 0.0380
4	0.6746 $\pm$ 0.0132	0.7439 $\pm$ 0.0106	0.7979 $\pm$ 0.0150	0.7277 $\pm$ 0.0274	0.6773 $\pm$ 0.0233
5	0.7488 $\pm$ 0.0204	0.7916 $\pm$ 0.0315	0.8040 $\pm$ 0.0214	0.7498 $\pm$ 0.0304	0.6922 $\pm$ 0.0481
6	0.7490 $\pm$ 0.0375	0.7361 $\pm$ 0.0175	0.7900 $\pm$ 0.0193	0.7924 $\pm$ 0.0289	0.7466 $\pm$ 0.0511
7	0.7201 $\pm$ 0.0264	0.7176 $\pm$ 0.0514	0.7463 $\pm$ 0.0325	0.7558 $\pm$ 0.0308	0.7261 $\pm$ 0.0395
8	0.7472 $\pm$ 0.0260	0.6975 $\pm$ 0.0245	0.7585 $\pm$ 0.0178	0.7760 $\pm$ 0.0285	0.7837 $\pm$ 0.0342
9	0.7945 $\pm$ 0.0200	0.7119 $\pm$ 0.0227	0.7803 $\pm$ 0.0545	0.7759 $\pm$ 0.0214	0.7605 $\pm$ 0.0322
10	0.7733 $\pm$ 0.0457	0.7497 $\pm$ 0.0403	0.7653 $\pm$ 0.0354	0.7436 $\pm$ 0.0484	0.7833 $\pm$ 0.0530
11	0.7567 $\pm$ 0.0187	0.7760 $\pm$ 0.0343	0.7543 $\pm$ 0.0363	0.7496 $\pm$ 0.0163	0.7752 $\pm$ 0.0469
12	0.7550 $\pm$ 0.0113	0.7989 $\pm$ 0.0189	0.7704 $\pm$ 0.0260	0.7683 $\pm$ 0.0152	0.7662 $\pm$ 0.0374

Table 13: Layer-wise diagnosis probing across DAPF, transcript-only BERT-CLS, and prompt-as-input BERT-CLS+Prompt. Values are macro-F1 scores averaged across five random seeds. For DAPF, [MASK] is the supervised masked-token diagnosis prediction position. For BERT-CLS+Prompt, the auxiliary [MASK] token is present in the input but is not the supervised prediction site; prediction is made through a standard [CLS] classification head.

Statistic	Mean Signed Attribution	Mean Abs. Attribution
Mean	-0.0015	0.0724
Standard Dev.	0.0608	0.0637
Minimum	-0.5181	0.0177
25th Percentile	-0.0174	0.0427
Median	0.0000	0.0540
75th Percentile	0.0147	0.0753
Maximum	0.2540	0.5775

Table 14: Summary statistics for aggregated DAPF token attributions ($n = 163$).

Component	Shared setting
Backbone	BERT-base-uncased
Training data	CCC + ADReSS train
Test data	ADReSS test
Test size	48 transcripts
Epochs	10
Batch size	4
Class weighting	Enabled
Random seeds	0, 1, 2, 3, 4

Table 15: Experimental settings shared by DAPF-BERT, BERT-CLS, and BERT-CLS+Prompt.

fragments such as *ing* and *tin* suggest sensitivity to tokenization conventions. Semantically related forms do not always receive the same direction: for example, *dry* has a positive mean attribution, whereas *drying* has a negative one; similarly, *reach* is positive while *reaching* is negative. These differences suggest that individual token directions should not be treated as stable linguistic evidence. Overall, Partition SHAP identifies the same broad types of cues seen with Integrated Gradients, but the two methods do not consistently agree on which

Metric	Value
IG top-20 tokens with matching SHAP direction	11 / 20
Spearman correlation	0.362

Table 16: Agreement between Integrated Gradients and Partition SHAP token attributions.

individual tokens are most important.

To conclude, Partition SHAP highlighted the same broad categories of evidence identified by Integrated Gradients, including picture-description vocabulary, discourse markers, and transcription artifacts. However, individual words often received different attribution magnitudes or even opposite attribution directions, indicating that token-level explanations remain sensitive to the choice of attribution method.

I Faithfulness Details

Table 18 extends Table 9 by reporting standard deviations across random seeds.

Table 19 reports faithfulness using tokens ranked by absolute attribution magnitude. Negative probability drops can thus occur when the highest-magnitude tokens mostly push the model away from DEMENTIA, and we observe this pattern for all three models. DAPF shows negative deletion and insertion effects overall. BERT-CLS shows increasingly negative deletion effects, and its insertion effect becomes negative at 50%. BERT-CLS+Prompt shows the strongest negative deletion effects, indicating that many high-magnitude tokens provide evidence against DEMENTIA.

Positive Mean SHAP Attribution					Negative Mean SHAP Attribution				
Token	Freq.	Trans.	Mean	Mean Abs.	Token	Freq.	Trans.	Mean	Mean Abs.
dry	12	9	0.0448	0.0464	flow	23	18	-0.0307	0.0349
ing	21	18	0.0428	0.0477	cookies	30	21	-0.0225	0.0267
fall	22	19	0.0323	0.0335	cookie	76	36	-0.0222	0.0252
reach	11	8	0.0278	0.0278	open	15	12	-0.0167	0.0172
get	23	13	0.0243	0.0284	guess	10	9	-0.0164	0.0218
oh	19	9	0.0176	0.0194	drying	20	16	-0.0155	0.0161
well	30	22	0.0162	0.0213	dish	12	10	-0.0151	0.0158
yeah	12	7	0.0157	0.0160	standing	20	18	-0.0141	0.0161
but	20	16	0.0151	0.0169	reaching	19	15	-0.0132	0.0139
got	15	8	0.0119	0.0120	plate	11	7	-0.0114	0.0120
tin	14	8	0.0109	0.0109	from	14	10	-0.0114	0.0137
and	240	45	0.0104	0.0138	think	12	11	-0.0105	0.0158
some	20	10	0.0096	0.0105	getting	12	11	-0.0097	0.0117
gonna	14	10	0.0086	0.0162	is	169	41	-0.0092	0.0132
him	15	12	0.0084	0.0102	stool	61	40	-0.0083	0.0134

Table 17: Tokens with the highest positive and negative mean Partition SHAP attributions across the ADReSS test set after applying the same token-reporting and normalization procedure used for Integrated Gradients. Only tokens occurring at least 10 times and in at least five transcripts are included. *Freq.* is the total number of occurrences, *Trans.* is the number of transcripts containing the token, *Mean* is the mean signed attribution to the DEMENTIA output, and *Mean Abs.* is the mean absolute attribution.

Model	Test	5%	20%	50%
DAPF	Deletion	0.0080	0.0055	-0.0318
	Drop	± 0.0155	± 0.0132	± 0.0144
DAPF	Insertion	-0.0187	-0.0352	-0.0916
	Recover	± 0.0192	± 0.0395	± 0.0612
BERT-CLS	Deletion	0.0259	0.0142	0.0067
	Drop	± 0.0287	± 0.0331	± 0.0280
BERT-CLS	Insertion	0.0369	0.0813	0.1223
	Recover	± 0.0226	± 0.0392	± 0.0558
BERT-CLS	Deletion	0.0302	0.0426	0.0259
+Prompt	Drop	± 0.0277	± 0.0409	± 0.0444
BERT-CLS	Insertion	0.0279	0.0787	0.1225
+Prompt	Recover	± 0.0185	± 0.0291	± 0.0417

Table 18: Positive-ranking faithfulness comparison using tokens ranked by positive attribution toward DEMENTIA. Positive values indicate behavior consistent with dementia-supporting evidence. Results are shown for 5%, 20%, and 50% perturbation fractions.

Model	Test	5%	20%	50%
DAPF	Deletion	-0.0511	-0.0939	-0.1799
	Drop	± 0.0471	± 0.0499	± 0.0483
DAPF	Insertion	-0.0709	-0.1902	-0.3453
	Recover	± 0.0607	± 0.1175	± 0.1019
BERT-CLS	Deletion	-0.0166	-0.1318	-0.3318
	Drop	± 0.0267	± 0.0524	± 0.1184
BERT-CLS	Insertion	0.0407	0.0082	-0.1940
	Recover	± 0.0247	± 0.0353	± 0.1054
BERT-CLS+Prompt	Deletion	-0.0997	-0.2207	-0.3883
	Drop	± 0.0775	± 0.0907	± 0.1016
BERT-CLS+Prompt	Insertion	0.0008	-0.0189	-0.2402
	Recover	± 0.0102	± 0.0395	± 0.0823

Table 19: Absolute-ranking faithfulness. Results should not be interpreted as target-specific rationales.